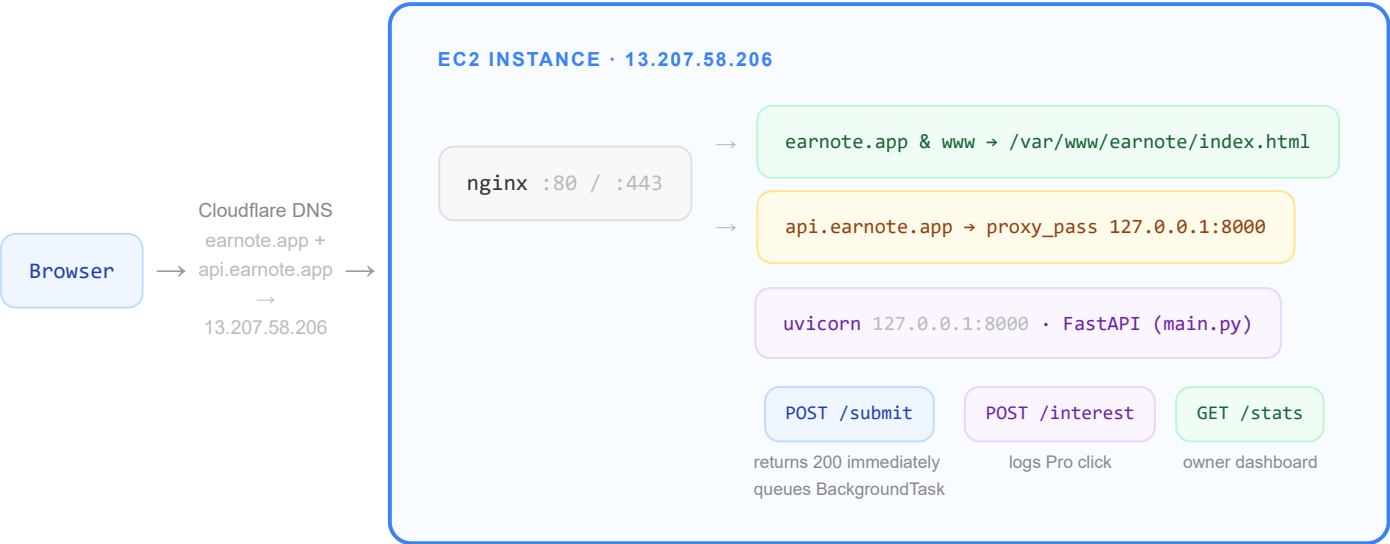


Earnote — System Architecture

Single EC2 box · Ubuntu · earnote.app · June 2026

REQUEST FLOW



BACKGROUND PIPELINE (triggered by POST /submit, runs after HTTP response)



Steps 2, 3, 4 are external API calls. yt-dlp uses youtube-cookies.txt to bypass YouTube bot-checks on EC2 IPs.

EC2 FILESYSTEM

/opt/earnote/	/var/www/earnote/
- main.py FastAPI app + pipeline functions - readback.db SQLite — jobs & interest tables - earnote.env API keys · chmod 600 · not in git - youtube-cookies.txt Edge cookies · chmod 600 - venv/ Python virtualenv + all pip packages	index.html entire frontend — one static file
	/etc/nginx/sites-enabled/
	- earnote-frontend serves earnote.app + www - earnote-api proxies api.earnote.app → :8000
	/etc/systemd/system/
	earnote.service keeps uvicorn alive, restarts on crash
	/etc/letsencrypt/live/earnote.app/

└─ fullchain.pem +
privkey.pem

TLS cert · covers 3 names ·
auto-renews

DATABASE /opt/earnote/readback.db (SQLite — single file, no separate server)

jobs		
id	INTEGER PK	autoincrement
email	TEXT	Lowercased on insert
url	TEXT	YouTube or audio URL
status	TEXT	queued → done / error: msg
created	TEXT	UTC ISO timestamp
Free-tier cap: COUNT(*) WHERE email=? ≥ 3 → 429 Powers: summaries_run, emails_at_cap in /stats		

interest		
id	INTEGER PK	autoincrement
email	TEXT	who clicked the \$9 button
plan	TEXT	"pro_9" (no charge taken)
created	TEXT	UTC ISO timestamp
Powers: pro_clicks in /stats — your conversion signal		

Important: The database is a single file on the EC2 instance. If the instance is terminated without a backup, all data is lost. To back up: `scp -i summary-website.pem ubuntu@13.207.58.206:/opt/earnote/readback.db ./readback.db`